Development of Online Quiz Module

Business Requirements Specification

Version 1.0

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Document Review and Acceptance

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Version	Name	Revision description	Date
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1. Introduction

1.1 Purpose

The purpose of this document is to define the functional requirements of a scalable and robust quiz application designed to assess students' knowledge in various aspects of robotics. The system will enable evaluation at individual, school, and district levels and generate meaningful reports for decision-making.

1.2 Scope

The Robotics Quiz Application will provide a platform for conducting quizzes, evaluating student performance, and generating analytics at multiple administrative levels. It will support multiple user roles including students, teachers, school administrators, district administrators, and super administrators. The system will ensure scalability, security, and reliability for large-scale usage.

1.3 Definitions and Acronyms

- **SRS**: Software Requirements Specification
- **RBAC**: Role-Based Access Control
- **Admin**: Administrator
- **MCQ**: Multiple Choice Question

2. Overall Description

2.1 Product Perspective

The system is a centralized web and mobile-based application designed to manage robotics assessments across multiple institutions. It integrates quiz management, evaluation, reporting, and analytics into a unified platform.

2.2 Product Functions

The system will:

- Manage users and roles
- Maintain student, school, and district data
- Create and manage question banks
- Conduct quizzes and evaluations
- Generate reports and analytics
- Ensure secure and scalable operations

2.3 User Classes and Characteristics

- **Students**: Attempt quizzes and view results
- **Teachers/Coordinators**: Manage quizzes and monitor student performance

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- **School Admins**: Manage school-level data and reports
- **District Admins**: Monitor district-wide performance
- **Super Admins**: Full system control and configuration

3. Functional Requirements

3.1 User Management Module

- The system shall allow registration of multiple user types.
- The system shall support login, logout, and password reset functionality.
- The system shall enforce role-based access control.
- The system shall allow administrators to activate or deactivate user accounts.
- The system shall maintain user status records.

3.2 Student Profile Module

- The system shall assign a unique ID to each student.
- The system shall store student details including school and district mapping.
- The system shall maintain enrolment status and learning level.
- The system shall track quiz attempts and performance history.
- The system shall provide progress tracking over time.

3.3 School Management Module

- The system shall maintain school details and classifications.
- The system shall map students and teachers to schools.
- The system shall support school-level quiz assignments.
- The system shall generate school-level performance reports.

3.4 District Management Module

- The system shall maintain district data and school mappings.
- The system shall manage district-level administrators.
- The system shall track district-wide participation.
- The system shall generate district-level analytics and reports.

3.5 Question Bank Management Module

- The system shall support creation and editing of questions.
- The system shall allow bulk upload of questions.
- The system shall support multiple question types.

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- The system shall categorize and tag questions by topic.
- The system shall store answers, explanations, and marks.
- The system shall implement a review and approval workflow.

3.6 Quiz Creation Module

- The system shall allow automatic and manual quiz creation.
- The system shall support topic-wise question selection.
- The system shall allow configuration of quiz parameters such as duration and marks.
- The system shall support randomization of questions and options.
- The system shall allow negative marking and passing criteria.

3.7 Quiz Assignment and Scheduling Module

- The system shall allow scheduling of quizzes with defined time windows.
- The system shall define attempt limits for quizzes.
- The system shall support retake policies.

3.8 Quiz Delivery Module

- The system shall provide a responsive interface for quiz delivery.
- The system shall display instructions and timer during quizzes.
- The system shall support autosave and resume functionality.
- The system shall allow navigation between questions.
- The system shall ensure reliable submission handling.

3.9 Evaluation and Scoring Module

- The system shall automatically evaluate objective questions.
- The system shall validate answers and assign marks.
- The system shall calculate topic-wise and overall scores.
- The system shall generate grades, ranks, and pass/fail results.

3.10 Assessment Report Module

- The system shall generate individual student reports.
- The system shall generate school-level performance reports.
- The system shall generate district-level consolidated reports.
- The system shall provide topic-wise analysis and participation data.

3.11 Analytics Dashboard Module

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- The system shall display quiz statistics and participation data.
- The system shall show average scores across schools and districts.
- The system shall provide topic-wise performance insights.
- The system shall display completion and dropout rates.

3.12 Admin Management Module

- The system shall provide a centralized admin dashboard.
- The system shall allow user approvals and role assignments.
- The system shall support onboarding of schools and districts.
- The system shall enable quiz monitoring and moderation.
- The system shall allow system configuration and overrides.

3.13 Security and Access Control Module

- The system shall implement secure authentication and authorization.
- The system shall store passwords securely using hashing.
- The system shall manage sessions securely.
- The system shall prevent unauthorized access.
- The system shall encrypt sensitive data.
- The system shall prevent misuse and duplicate attempts.

3.14 Scalability and Performance Module

- The system shall support high concurrent user load.
- The system shall ensure fast response times.
- The system shall optimize database operations.
- The system shall implement caching and load balancing.
- The system shall ensure high availability.

3.15 Backup and Recovery Module

- The system shall perform periodic backups.
- The system shall store backups of all critical data.
- The system shall provide restore mechanisms.
- The system shall support disaster recovery processes.

4. Non-Functional Requirements

4.1 Performance

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- The system shall handle thousands of concurrent users efficiently.

4.2 Security

- The system shall ensure data privacy and secure access.

4.3 Reliability

- The system shall ensure minimal downtime and data loss.

4.4 Scalability

- The system shall scale horizontally to support growth.

4.5 Usability

- The system shall provide an intuitive and user-friendly interface.

5. Data Requirements

- The system shall store structured data for users, student UniqueId, schools, districts, quizzes, and results.
- The system shall ensure data consistency and integrity.
- The system shall maintain relationships between entities (student–school–district).
- The system shall support large-scale data storage (100,000+ records).

6. UI Component Library

The system shall utilize categorized UI components including input controls, navigation elements, data visualization components, feedback mechanisms, and interaction overlays to ensure consistency, usability, and responsiveness across the application.

Category	Components
Form & Input Components	Button, Input, Select / Dropdown, Checkbox / Radio, Date Picker
Layout & Navigation Components	Sidebar, Navbar, Tabs, Accordion, Card
Data Display Components	Table / Data Grid, Pagination, Chart Container, Progress Bar
Feedback & Notification Components	Toast, Alert, Tooltip
Overlay & Interaction Components	Dialog / Modal
Utility Components (Optional)	Progress Bar, Tooltip

7. Technology Architecture

7.1 Frontend Stack:

- React 18.3.1 – Core framework.
- JavaScript/TypeScript – Type safety and developer experience.
- Tailwind CSS v4.0 – Utility-first styling.

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- React Router – Client-side routing.
- React Query – Data management.
- Lucide React – Icon library.
- Recharts – Data visualization.

7.2 Backend Stack:

- Ruby – Business logic and RESTful API services.
- PostgreSQL with Sequel ORM – Database operations, supporting geospatial data.

7.3 Integration Points:

- Database API: RESTful endpoints for school, college, activity, and geospatial data.
- Authentication Service: JWT-based with email OTP fallback for secure staff access.
- File Upload Service: Cloud storage for images, documents, and PDF reports.

7.4 Features:

- Responsive Design: Mobile-first with full device compatibility.
- Component-Based Architecture: Reusable, modular components for maintainability.
- Modern UI: Clean, professional design styled
- Performance Optimized: Page load < 3s, PDF export < 5s.
- Accessibility: WCAG 2.1 Level AA compliant.
- Scalability: Supports high data volumes and user traffic.

8. Constraints

None

9. Use Case Scenarios

9.1 Student Takes Quiz

UseCase Name	**Student Takes Quiz**
Actor	Student
Description	This use case describes how a student logs in, accesses an assigned quiz, attempts it, and submits it for evaluation.
Preconditions	The student must be registered in the system and assigned an active quiz.
Postconditions	The quiz is submitted successfully, responses are stored, and the result is generated.
Main Flow	1. Student logs into the system. 2. Student views available quizzes. 3. Student selects an assigned quiz. 4. System displays quiz instructions. 5. Student starts the quiz. 6. Student answers questions. 7. System autosaves responses during

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UseCase Name	**Student Takes Quiz**
	the attempt. 8. Student reviews answers. 9. Student submits the quiz. 10. System evaluates the responses and stores the result.
Alternate Flow	If the network fails during the attempt, the system saves existing responses and allows the student to resume later within the permitted time window.

9.2 Admin Creates Quiz

UseCase Name	**Admin Creates Quiz**
Actor	Teacher / Coordinator / School Admin / Super Admin
Description	This use case describes how an authorized user creates a new quiz using the question bank.
Preconditions	The actor must have permission to create quizzes and the question bank must contain approved questions.
Postconditions	A new quiz is created and stored in the system.
Main Flow	1. Admin logs into the system. 2. Admin navigates to the quiz creation module. 3. Admin enters quiz details such as title, instructions, duration, and total marks. 4. Admin selects questions from the question bank or chooses auto-generation. 5. Admin sets rules such as number of questions, randomization, negative marking, and passing criteria. 6. Admin saves the quiz.
Alternate Flow	If insufficient questions are available for the selected topic, the system displays an alert and requests modification of quiz settings.

9.3 Admin Assigns and Schedules Quiz

UseCase Name	**Admin Assigns and Schedules Quiz**
Actor	Teacher / School Admin / District Admin / Super Admin
Description	This use case describes how an authorized user assigns a quiz to students and schedules its availability.
Preconditions	The quiz must already exist in the system.
Postconditions	The quiz is assigned to selected students, classes, schools, or districts with defined schedule settings.
Main Flow	1. Admin selects an existing quiz. 2. Admin chooses the target group such as class, school, or district. 3. Admin sets start date, end date, availability

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UseCase Name	**Admin Assigns and Schedules Quiz**
	window, and attempt limits. 4. Admin defines retake policy if applicable. 5. Admin confirms the assignment. 6. System publishes the quiz as scheduled.
Alternate Flow	If the selected schedule conflicts with an existing active schedule, the system warns the admin and requests correction.

9.4 System Evaluates Quiz

UseCase Name	**System Evaluates Quiz**
Actor	System
Description	This use case describes how the system automatically evaluates submitted quiz responses.
Preconditions	The student must have submitted a quiz attempt.
Postconditions	The quiz score, percentage, grade, and pass/fail result are calculated and stored.
Main Flow	1. System receives submitted responses. 2. System validates answers against correct answers stored in the question bank. 3. System calculates marks for each question. 4. System computes topic-wise scores. 5. System calculates total score, percentage, grade, and pass/fail status. 6. System stores the final result.
Alternate Flow	If a question is invalid or missing evaluation parameters, the system marks it for admin review and continues processing the remaining questions.

9.5 Student Views Result

UseCase Name	**Student Views Result**
Actor	Student
Description	This use case describes how a student views the result of a completed quiz.
Preconditions	The student must have completed and submitted the quiz, and evaluation must be available.
Postconditions	The student can view their quiz scorecard and performance summary.
Main Flow	1. Student logs into the system. 2. Student navigates to the results section. 3. Student selects a completed quiz. 4. System displays score, percentage, grade, and topic-wise performance.
Alternate Flow	If results are not yet published, the system displays a pending status message.

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9.6 School Admin Views School Report

UseCase Name	**School Admin Views School Report**
Actor	School Admin
Description	This use case describes how a school administrator views school-level quiz performance reports.
Preconditions	Quiz attempts must exist for students in the school.
Postconditions	School-level summary data is displayed to the school admin.
Main Flow	1. School admin logs into the system. 2. School admin opens the reports module. 3. School admin selects a quiz or reporting period. 4. System generates school-level statistics such as participation, average score, and topic-wise strengths and weaknesses. 5. School admin views or exports the report.
Alternate Flow	If there are no quiz attempts for the selected period, the system displays a no-data message.

9.7 District Admin Views District Report

Use Case Name	**District Admin Views District Report**
Actor	District Admin
Description	This use case describes how a district administrator views district-level consolidated reports.
Preconditions	Schools and quiz attempt data must be available under the district.
Postconditions	The district admin can view district-wide analytics and school comparisons.
Main Flow	1. District admin logs into the system. 2. District admin opens the analytics or reports module. 3. District admin selects district-wide reporting criteria. 4. System generates district-level summary including school participation, average scores, and comparative analysis. 5. District admin views or exports the report.
Alternate Flow	If one or more schools have incomplete data, the system flags missing records in the report.

9.8 Admin Manages Question Bank

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UseCase Name	**Admin Manages Question Bank**
Actor	Teacher / Coordinator / Super Admin
Description	This use case describes how an authorized user adds, edits, uploads, and reviews questions in the question bank.
Preconditions	The actor must have question management privileges.
Postconditions	New or updated questions are stored in the question bank and made available based on approval status.
Main Flow	1. Admin logs into the system. 2. Admin opens the question bank module. 3. Admin adds a new question or uploads multiple questions in bulk. 4. Admin specifies question type, topic, tags, correct answer, explanation, and marks. 5. Admin submits the question for review or approval. 6. System stores the question.
Alternate Flow	If uploaded questions contain invalid or missing fields, the system rejects those records and displays validation errors.

9.9 Super Admin Manages Users

UseCase Name	**Super Admin Manages Users**
Actor	Super Admin
Description	This use case describes how the super admin creates, updates, activates, deactivates, or assigns roles to users.
Preconditions	The super admin must be authenticated.
Postconditions	User records and permissions are updated successfully.
Main Flow	1. Super admin logs into the system. 2. Super admin opens the user management module. 3. Super admin searches for a user or creates a new user. 4. Super admin assigns or updates role and status. 5. System saves the changes and updates access permissions.
Alternate Flow	If the role assignment violates system rules, the system denies the action and displays an error message.

9.10 System Performs Backup and Recovery

UseCase Name	**System Performs Backup and Recovery**
Actor	System / Super Admin

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UseCase Name	**System Performs Backup and Recovery**
Description	This use case describes how the system performs periodic backups and how an admin restores data when needed.
Preconditions	Backup configuration must be enabled.
Postconditions	Data is backed up successfully or restored from a valid backup.
Main Flow	1. System initiates scheduled backup. 2. System stores backup of quizzes, users, questions, and responses. 3. Super admin requests restore when needed. 4. System verifies backup integrity. 5. System restores the selected data set.
Alternate Flow	If the backup file is corrupted or unavailable, the system reports restoration failure and logs the issue.

10. User Interface Screens

Mention key screens:

- Login Screen
- Dashboard
- Quiz Screen
- Result Screen
- Admin Panel
- Reports Dashboard

11. Edge Cases Summary

The system shall handle various edge cases to ensure reliability, data integrity, and a seamless user experience. Key edge cases include:

- **User Management:**
  Duplicate registrations, invalid role assignments, and login attempts by deactivated users shall be properly handled.
- **Student Data:**
  Missing or incorrect student–school–district mappings and duplicate student IDs shall be prevented or flagged.
- **Question Bank:**
  Invalid question formats, missing correct answers, duplicate questions, and errors in bulk uploads shall be detected and rejected.
- **Quiz Creation:**
  Quizzes with insufficient questions, invalid duration, incorrect marking rules, or empty configurations shall not be allowed.

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- **Quiz Scheduling:**
  Invalid date ranges, overlapping schedules, and assignments to inactive or non-existent users shall be restricted.
- **Quiz Attempt:**
  Network failures, session interruptions, browser refresh, multiple device access, and timer expiry shall be handled with autosave and controlled recovery.
- **Evaluation:**
  Missing answer keys, partial submissions, duplicate submissions, and tie scores shall be managed according to defined rules.
- **Reporting:**
  Scenarios with no data, incomplete data, or changes in student mapping shall be handled without generating misleading reports.
- **Admin Operations:**
  Deletion of dependent data, invalid overrides, and accidental changes to active users or quizzes shall be prevented or warned.
- **Security:**
  Unauthorized access, repeated failed logins, and attempt manipulation shall be detected and restricted.
- **System & Backup:**
  Backup failures, corrupted data, partial restores, and unexpected shutdowns shall be handled with logging and recovery mechanisms.

This SRS document defines the requirements for the Robotics Quiz Application, including user management, quiz operations, evaluation, and reporting functionalities. It also addresses system performance, security, and scalability aspects. These requirements provide a clear foundation for the design and development of a reliable and efficient assessment platform.

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